Yassine Oubaid

PhD in Condensed Matter Physics
Université Paris–Saclay

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Research Interests

- Strongly correlated electron systems and phase transitions
- Unconventional superconductors
- Exotic magnetism
- Synchrotron and neutron scattering techniques
- Optical spectroscopy (infrared/THz)
- Measurements under extreme conditions (High pressure / low temperature)

Education

2023–present **PhD in Condensed Matter Physics** – Université Paris–Saclay | SOLEIL Synchrotron | Solid state physics laboratory (LPS) *(25 February 2026)*
(International ranking: 6th worldwide in Physics — Shanghai Ranking 2025)
Thesis: Study of the interplay of magnetism, structure, and electronic properties in the Iron-Based spin ladder $BaFe_2S_3$.
Advisors: Pascale FOURY-LEYLEKIAN, Director of Solid state physics laboratory (LPS), Victor BALÉDENT, Professor, Pierre FERTEY, head of CRISTAL beamline, Marine VERSEILS, Beam scientist (AILES beamline).

2021–2022 **Engineering/MSc (M2)** – Ecole Centrale Lyon (ECL), France
General engineering curriculum in Ecole Centrale Lyon with a specialisation in nanoscience and a Master in nanoscience the same year (Master Nanoscale engineering (NSE) : ECL, INSA Lyon, Université de Lyon (ULCB)).

2019–2021 **Engineering Diploma** – Ecole Centrale Casablanca, Morocco
General engineering student with a nanoscience specialisation during the second year.
Excellence Scholarship : Bourse fondation OCP

2017–2019 **Preparatory Classes (PSI)** – Lycée d'excellence de Benguerir (LYDEX), Morocco
Ranked 30th nationwide (top 3% of the accepted candidates)

Research and Professional Experience

2023–present **PhD Student – Iron-based superconductor $BaFe_2S_3$**
(Université Paris–Saclay | Synchrotron SOLEIL | LPS, France)
Study of the interplay of magnetism, structure, and electronic properties in the iron-based spin ladder $BaFe_2S_3$.

2022 **Research Internship – Topological superconductivity in hybrid nanowires**
(CNRS (Institut Néel) | CEA (IRIG), Grenoble, France)
Tailoring superconductivity in core-shell InAs–ZnTe nanowires.

2021 **Research Internship – 2D materials growth**
(Ecole Centrale Casablanca | UM6P Benguerir, Morocco)
Contribution to a review paper on 2D materials, focusing on CVD (chemical vapor deposition) growth for FET (field-effect transistor) applications.

Experiments at Large Facilities

- **AILES beamline, Synchrotron SOLEIL (Paris):** infrared spectroscopy in : THz, Far-infrared, Mid-infrared, Near-infrared. (low temperature/high pressure).
- **CRISTAL beamline, Synchrotron SOLEIL (Paris):** Single crystal/powder X-ray diffraction. (low temperature/high pressure).
- **thALES, IN8 instruments, Institut Laue–Langevin (ILL) (Grenoble):** Inelastic neutron scattering.
- **D1B,D20 instruments, Institut Laue–Langevin (ILL) (Grenoble):** powder neutron diffraction under high pressure and low temperature.
- **D9,D23 instruments, Institut Laue–Langevin (Grenoble):** single-crystal neutron diffraction at low temperature and high magnetic field.
- **ESRF/ILL/ALBA Synchrotron:** Training experiments during HERCULES School.

Conferences

Jul 2025 **International conference: JEMS-2025** – Frankfurt, Germany, **Invited speaker**.
Jul 2025 **International conference: SPSSM-2025** – Nantes, France, **Oral presentation**.
2025 **GDR Meetic (CNRS)** – Paris, France, Thematic: New quantum states, **Oral presentation**.
2024 **Neutron Scattering Days (JDN2024)** – Île de Porquerolles, France, **Oral presentation**.
2024 **GDR Meetic (CNRS)** – Strasbourg, France, Thematic: Magnetism, **Oral presentation**.
2024 **Science Day (SOLEIL Synchrotron) (journée sciences et technique)** – Saint-Aubin, France, **Oral presentation**.
2023 **Neutron Scattering Days (JDN2023)** – Erquy, France, **Poster presentation**.
2023 **GDR Meetic (CNRS)** – Bordeaux, France, Plenary meeting, **Poster presentation**.
Apr 2022 **International conference: Nanowires Week 2022** – Chamonix Mont-Blanc, France, Participant and member of the organising committee.

Schools and trainings

Feb 2024 **HERCULES European School on Large Scale Facilities** – Université Grenoble Alpes, Grenoble, France.
May 2024 **Topical School GDR MEETICC: Topology** – Aussois, France.
2024 **Workshop on FullProf Suite software** – Grenoble, France.
Nov 2023 **Ad Hoc workshop on Jana2020 software** – Prague, Czech Republic.
Jun 2023 **Summer School in Mathematical Crystallography** – Lorraine University, Nancy, France.
2023 **Python for Physics** – Sorbonne University, Paris, France.
Apr 2022 **PhD Summer School in Condensed Matter Physics** – Center for Quantum Devices, Copenhagen, Denmark.

Teaching and Outreach

- **International physics Olympiade (IPHO): Paris 2025** Team leader of Morocco.
- **Teaching Assistant (doctoral contract)** – Université Paris–Saclay (Sept 2023 – Jan 2025)
  • Led experimental physics sessions (TPs) for engineering students at Polytech Paris–Saclay covering wave physics, polarisation, interference and related topics.
  • Analysis and linear algebra to first- and second-year undergraduate students (L1,L2) at Université Paris–Saclay.
- **Science communication :** Organized scientific outreach activities for high school students, including interactive sessions on superconductivity and general science communication events.
- **Supervision**– Supervision of two undergraduate student interns : Roméo Morlevat-Grivot: Bachelor student intern (L3), Anass Chaabi: Master student intern (M1).

Selected Publications:

- **Y.Oubaid**, V. Balédent, O. Fabelo, L. Bocher, C. V. Colin, S. Chattopadhyay, E. Elkaim, M. Verseils, A. Forget, D. Colson, D. Bounoua, P. Fertey, P. Foury-Leylekian, *New insight on the properties of the superconducting iron spin ladder* $BaFe_2S_3$, Frontiers of Physics **21**(5), 055201 (2026). doi:10.15302/frontphys.2026.055201 (highlights of the French research federation for the neutron scattering community. 2FDN)
- **Y.Oubaid**, S. Deng, N. S. Dhami, M. Verseils, D. Bounoua, A. Forget, D. Colson, P. Foury-Leylekian, M.-B. Lepetit, V. Balédent, *Ground State of* $BaFe_2S_3$ *from Lattice and Spin Dynamics*,Preprint: arXiv:2602.16899 [cond-mat.str-el] (2026), submitted to *Physical Review B*.
- **Y.Oubaid**, P. Foury-Leylekian, N.S.Dhami, C. Ritter, C. V. Colin, A. Forget, J. P. Rueff, M. Verseils, P. Fertey, V. Balédent, *Dichotomy between localized and itinerant Fe* 3*d electrons across the pressure-induced superconducting state of the spin-ladder compound* $BaFe_2S_3$, in preparation to be submitted in Nature Physics (ongoing).
- W. G. Zheng, V. Balédent, **Y.Oubaid**, P. Auban-Senzier, C.V. Colin, F. Damay, C. Pasquier, A. Forget, D. Colson, J. P. Xu, W. Yin, P. Miao, P. Foury-Leylekian, *Study of the transport and magnetic properties of substituted* $Ba(Fe_{1-x}Ni_x)_2(Se_{1-y}Te_y)_3$, Physical Review B **109**(18), 184428 (2024). doi:10.1103/PhysRevB.109.184428

Additional Activities and Interests

- Sports: volleyball, football, chess.
- Science communication: actively participates in outreach events and enjoys sharing physics with broad audiences.

References

Available upon request.